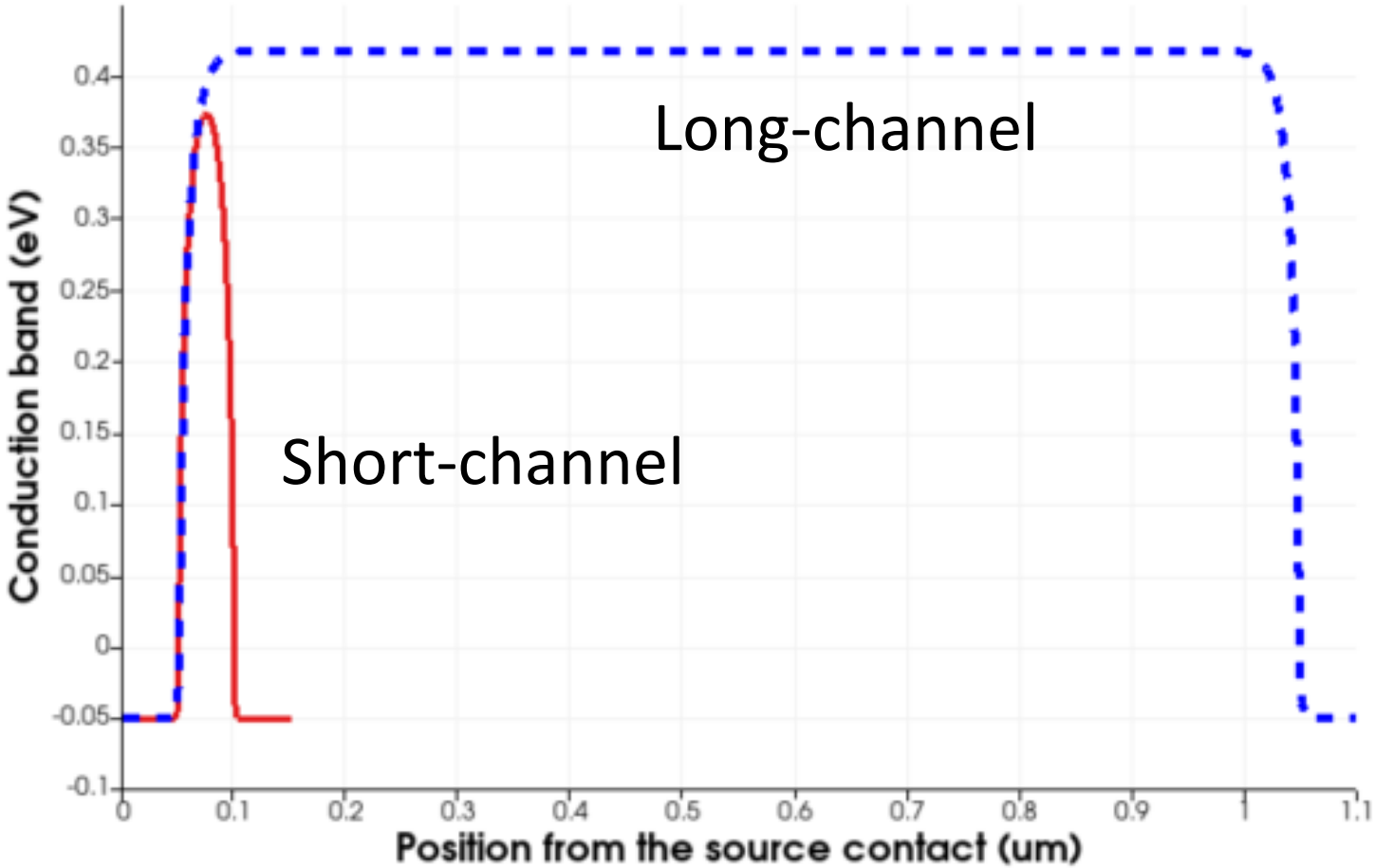
VLSI Devices
Lecture 16

Sung-Min Hong (smhong@gist.ac.kr)
Semiconductor Device Simulation Laboratory
Department of Electrical Engineering and Computer Science
Gwangju Institute of Science and Technology (GIST)

L16

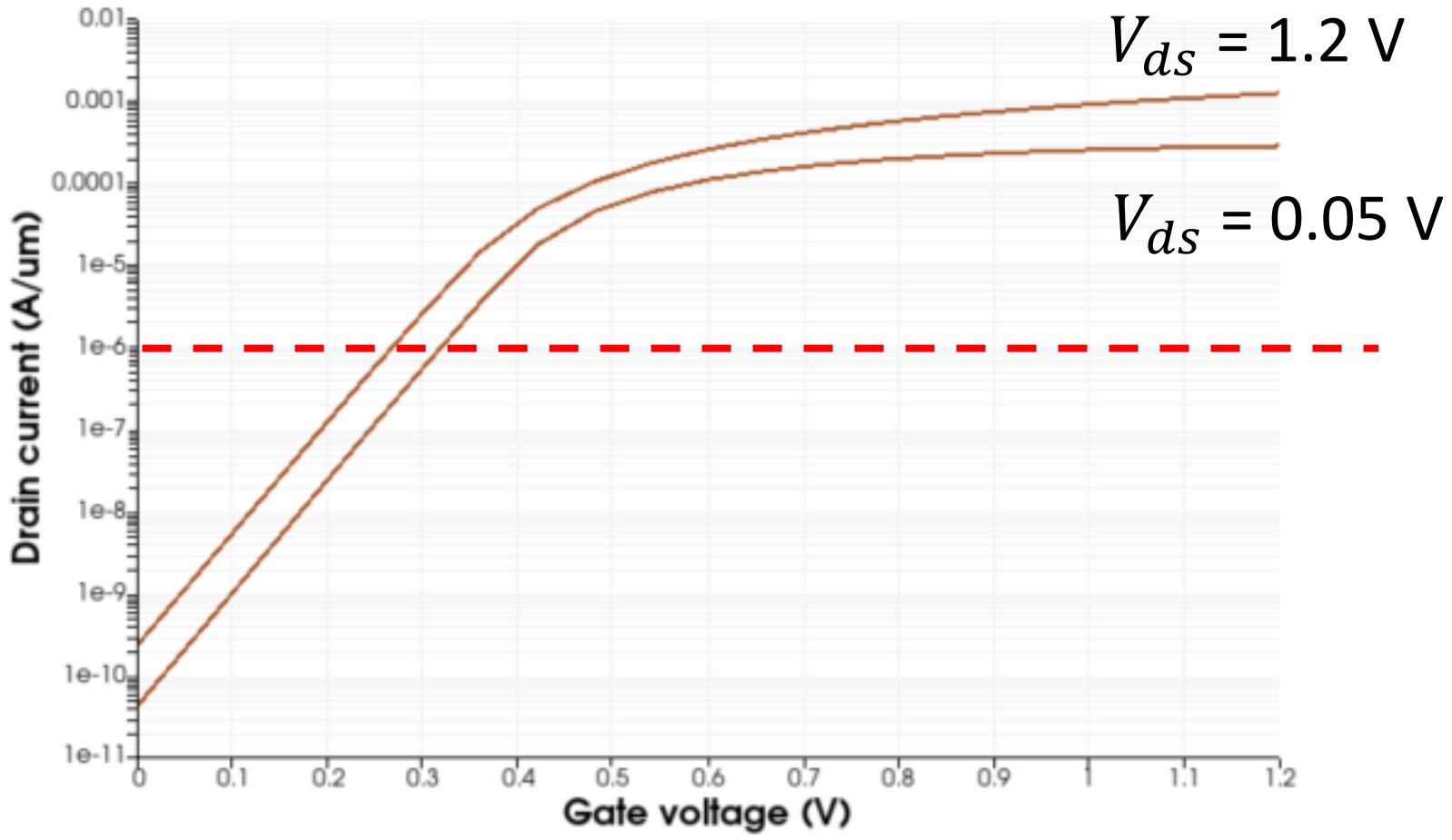
Charge sharing

- Even at V_{gs} = V_{ds} = 0 V, the conduction band profiles are different.



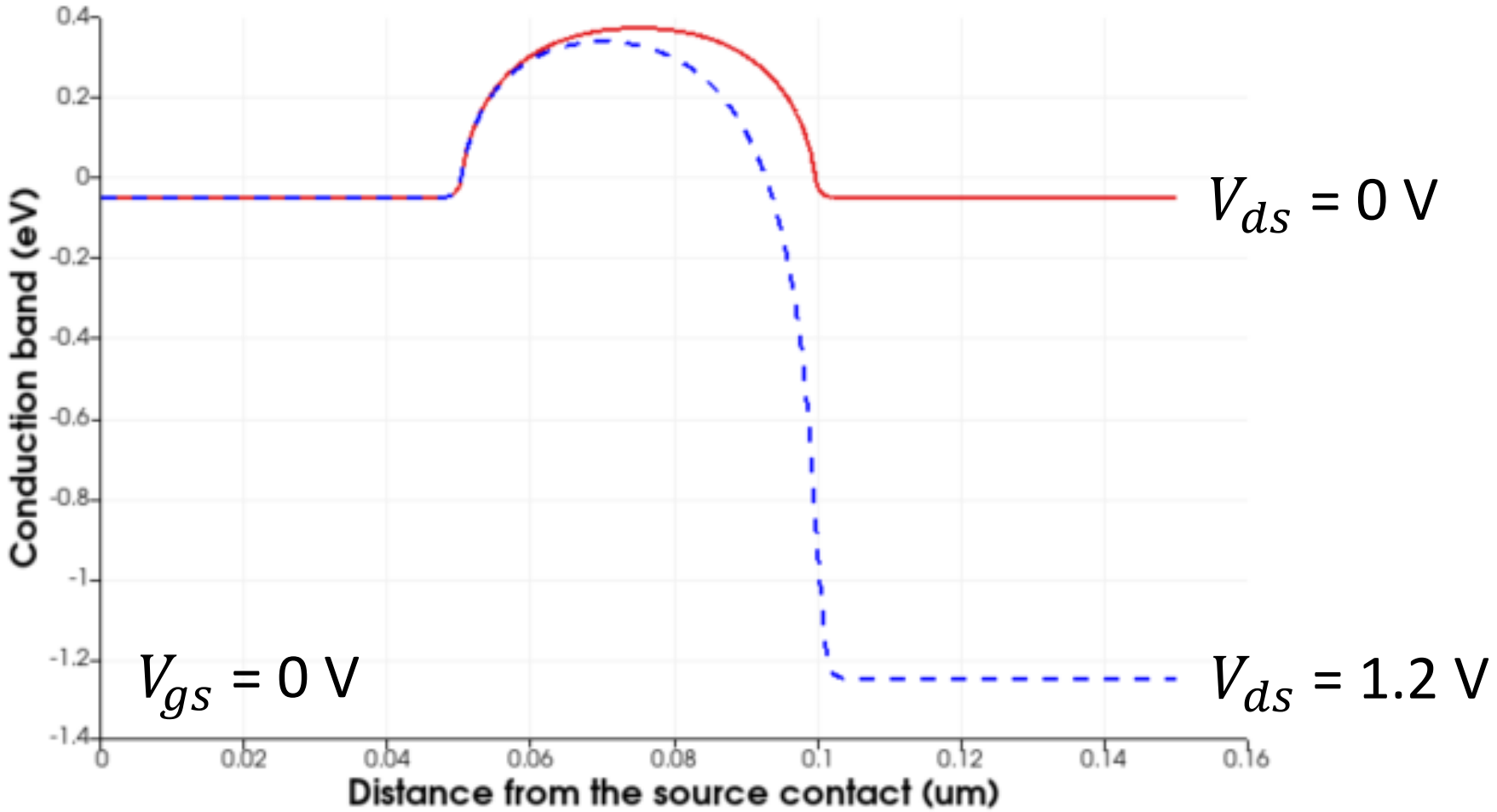
Drain-induced barrier lowering (DIBL)

- Much worse than the long-channel device
  - 45 mV/V @ I_D = 10^{-6} A/μm



Conduction band, again

- At a high V_{ds}, the energy barrier is further reduced.



Simplified geometry for an analytic solution

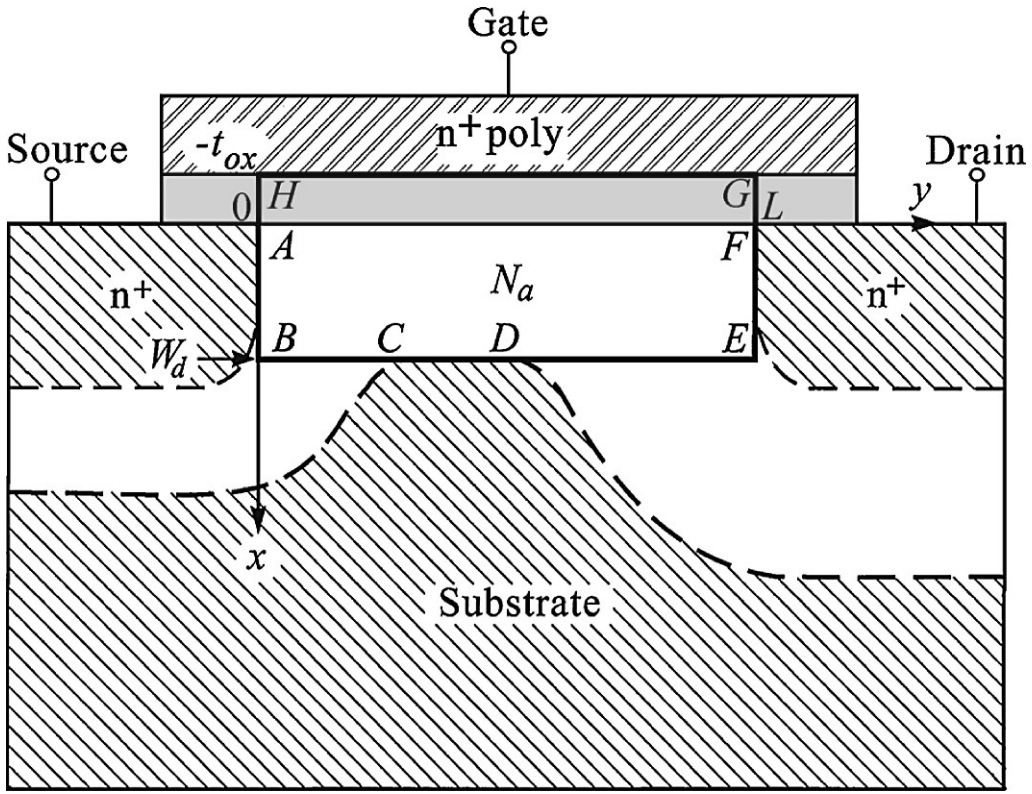
- Poisson equation
  - For AFGH

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0$$ Taur, Eq. (6.1)

  - For ABEF (Depleted)

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=\frac{qN_a}{\epsilon_{si}}$$

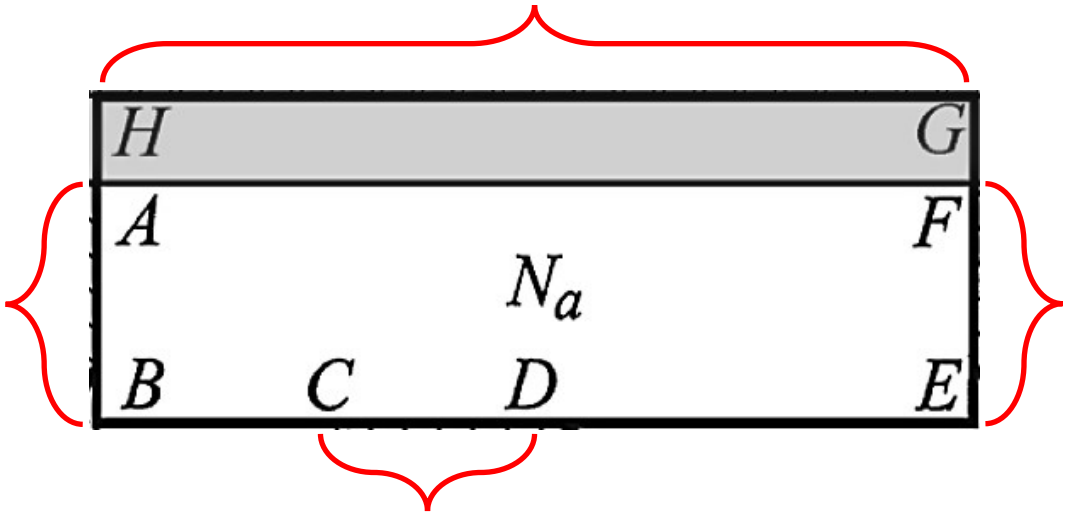
Taur, Eq. (6.2)



Simplified geometry (Taur, Fig. 6.5)

Boundary conditions

- Potential reference is $\phi(\infty, y) = -\phi_B$.
  - Along GH: $\phi = V_{gs} - V_{fb}$ Taur, Eq. (6.3)
  - Along AB: $\phi = \phi_{bi} \approx \frac{E_g}{2q} + \phi_B$ Taur, Eq. (6.4)
  - Along EF: $\phi = \phi_{bi} + V_{ds}$ Taur, Eq. (6.5)
  - Along CD: $\phi = 0$ Taur, Eq. (6.6)



Solution

- Poisson equation with boundary conditions
  - Try the following functions for the electrostatic potential.
  - For the oxide region,

$$\phi_1(x,y) = v_1(x,y) + u_{L1}(x,y) + u_{R1}(x,y)$$ Taur, Eq. (6.7)

  - For the silicon region,

$$\phi_2(x,y) = v_2(x,y) + u_{L2}(x,y) + u_{R2}(x,y)$$ Taur, Eq. (6.8)

Poisson equation with top/bottomr b.c. (Long-channel)

Laplace equation to match left b.c.

Laplace equation to match right b.c.

Solution, $v(x, y)$

- Actually, it is $v(x)$.
  - For the oxide region ($-t_{ox} \leq x \leq 0$),

$$v_1(x) = \phi_s - \frac{V_{gs} - V_{fb} - \phi_s}{t_{ox}} x$$

Taur, Eq. (6.9)

  - For the silicon region ($0 \leq x \leq W_d$),

$$v_2(x) = \phi_s \left(1 - \frac{x}{W_d}\right)^2$$

Taur, Eq. (6.10)

  - It is noted that

$$\phi_s = \frac{qN_a W_d^2}{2\epsilon_{si}}$$

Taur, Eq. (6.11)

Relation between $v_1(x)$ and $v_2(x)$

- At $x = 0$, the displacement must be continuous. (ϕ_s itself is already continuous.)
  - For the oxide region

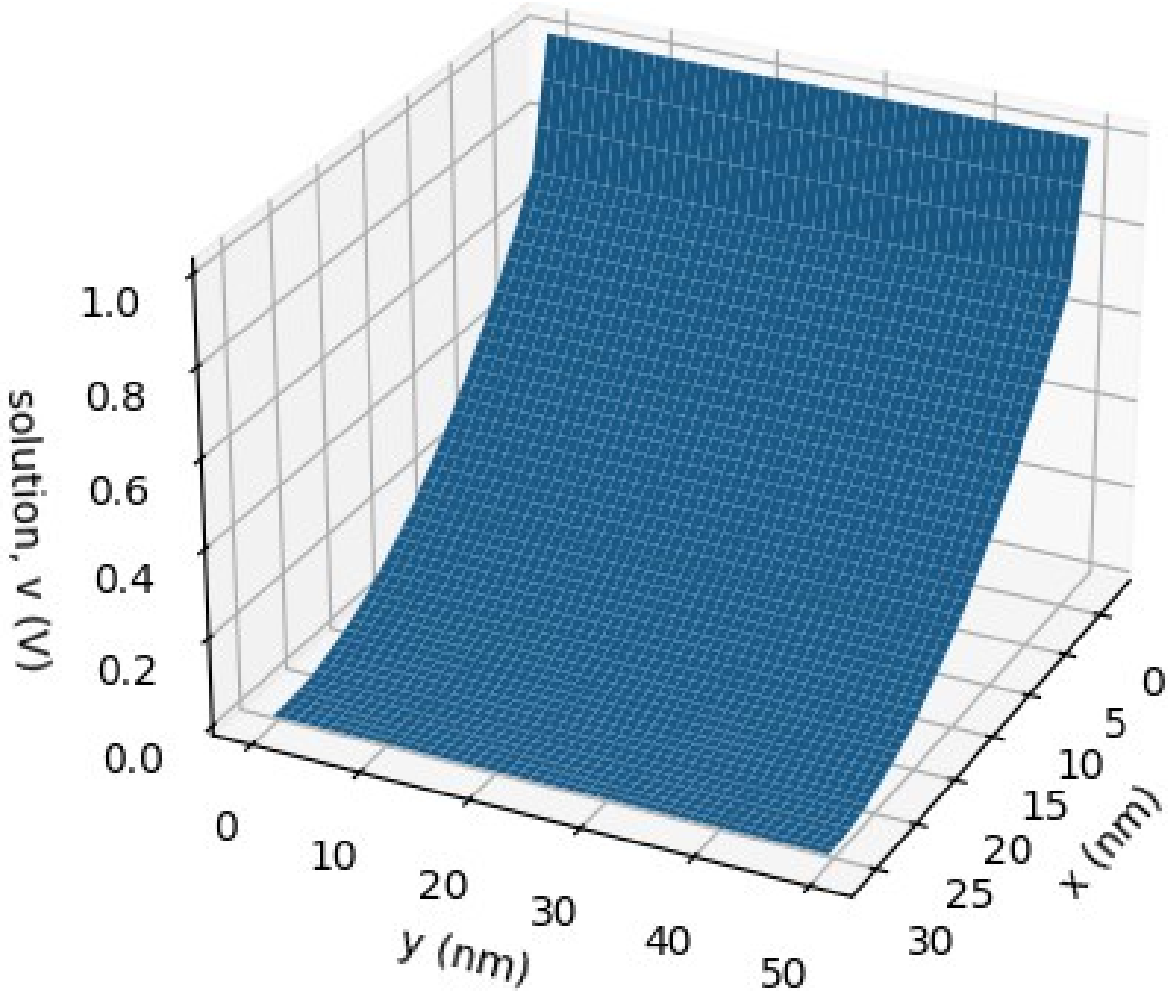
Taur, Eq. (6.12)

$$\epsilon_{ox}\frac{V_{gs} - V_{fb} - \phi_s}{t_{ox}} = \epsilon_{si}\frac{2\phi_s}{W_d} = \sqrt{2\epsilon_{si} q N_a \phi_s}$$

  - Example)
  - For our example, V_{fb} is -1.01 V, t_{ox} is 2.0 nm, N_a is 10^{18} cm^{-3}.
  - At zero gate voltage, ϕ_s is about 0.726 V.
  - The depletion width is about 30 nm.
  - (ϕ_B is about 0.474 V.)

Long-channel solution, $v_1(x)$ and $v_2(x)$

- 50-nm-long channel

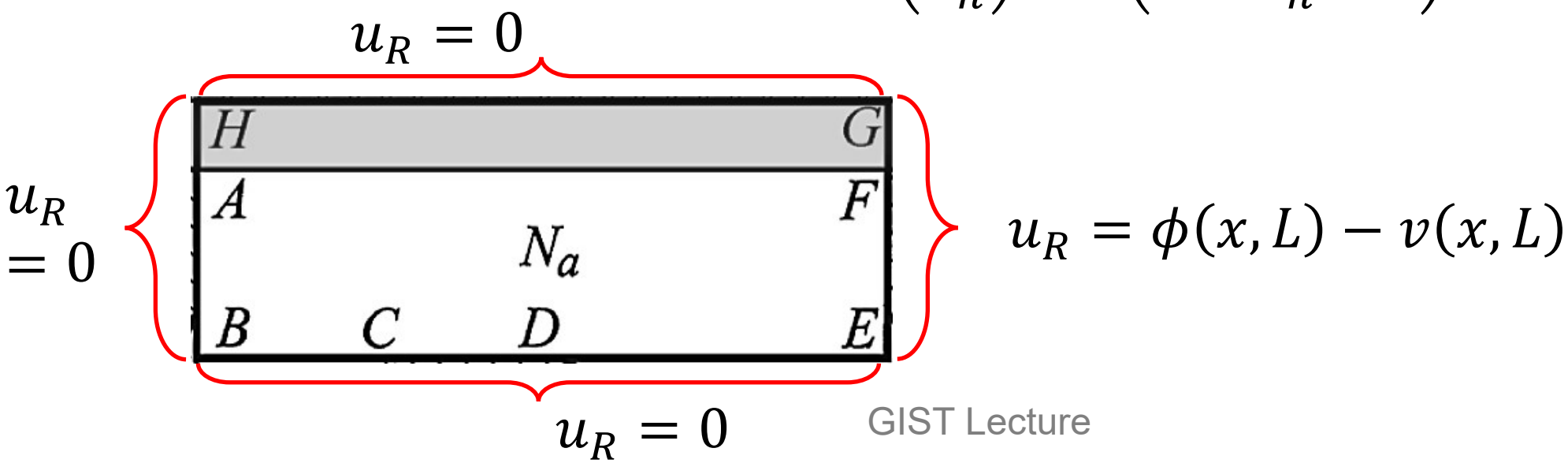


Modes for $u_{R1}(x,y)$ and $u_{R2}(x,y)$

- For three boundaries, it should vanish.

$$u_{R1,n}(x,y) = \sinh\left(\frac{\pi y}{\lambda_n}\right)\sin\left(\frac{\pi(x+t_{ox})}{\lambda_n}\right)$$ Taur, ~Eq. (6.14)

$$u_{R2,n}(x,y) = \sinh\left(\frac{\pi y}{\lambda_n}\right)\sin\left(\frac{\pi(W_d-x)}{\lambda_n}\right)$$ Taur, ~Eq. (6.16)



Series expansion of $u_R(x, y)$

- $u_R(x, L) = \phi(x, L) - v(x, L)$ can be expanded with coefficients c_n's.
  - Therefore,

$$u_{R1}(x, y) = \sum_{n=1}^{\infty} c_{1,n} \frac{\sinh\left(\frac{\pi y}{\lambda_n}\right)}{\sinh\left(\frac{\pi L}{\lambda_n}\right)} \sin\left(\frac{\pi(x + t_{ox})}{\lambda_n}\right)$$

Taur, Eq. (6.14)

$$u_{R2}(x, y) = \sum_{n=1}^{\infty} c_{2,n} \frac{\sinh\left(\frac{\pi y}{\lambda_n}\right)}{\sinh\left(\frac{\pi L}{\lambda_n}\right)} \sin\left(\frac{\pi(W_d - x)}{\lambda_n}\right)$$

Taur, Eq. (6.16)

Relation between $u_{R1}(x, y)$ and $u_{R2}(x, y)$

- At $x = 0$, the potential and the displacement must be continuous.

$$c_{1,n} \sin\left(\frac{\pi t_{ox}}{\lambda_n}\right) = c_{2,n} \sin\left(\frac{\pi W_d}{\lambda_n}\right)$$ Taur, ~Eq. (6.17)

$$c_{1,n} \epsilon_{ox} \cos\left(\frac{\pi t_{ox}}{\lambda_n}\right) = -c_{2,n} \epsilon_{si} \cos\left(\frac{\pi W_d}{\lambda_n}\right)$$ Taur, ~Eq. (6.18)

  - We don't know $c_{1,n}$, $c_{2,n}$, and λ_n.

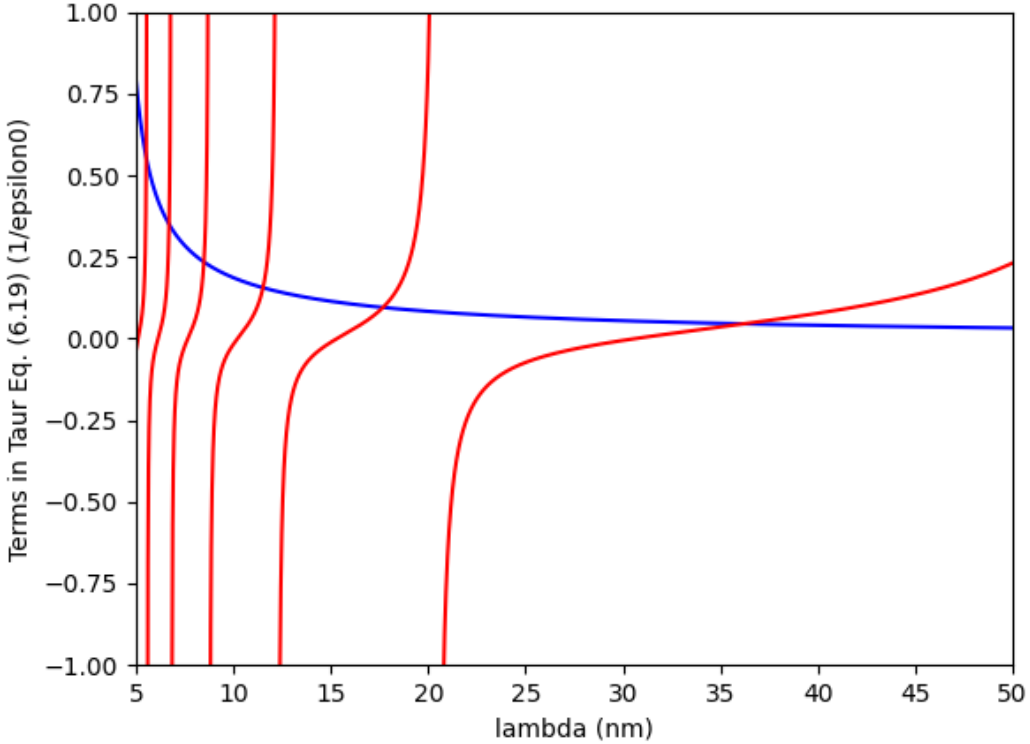
Equation for λ_n

- the potential and the displacement must be continuous.

$$\frac{1}{\epsilon_{ox}}\tan\left(\frac{\pi t_{ox}}{\lambda_n}\right) = -\frac{1}{\epsilon_{si}}\tan\left(\frac{\pi W_d}{\lambda_n}\right)$$

Taur, Eq. (6.19)

  - There are multiple solutions: $\lambda_1, \lambda_2, \ldots$
  - Example)



Full 2D potential solution

- the potential and the displacement must be continuous.

$$\phi_2(x,y) = \phi_s\left(1-\frac{x}{W_d}\right)^2 + \sum_{n=1}^{\infty}\frac{b_{2,n}\sinh\left(\frac{\pi(L-y)}{\lambda_n}\right)+c_{2,n}\sinh\left(\frac{\pi y}{\lambda_n}\right)}{\sinh\left(\frac{\pi L}{\lambda_n}\right)}\sin\left(\frac{\pi(W_d-x)}{\lambda_n}\right)$$

Taur, Eq. (6.20)

  - Coffecients ($b_{2,n}$ and $c_{2,n}$) are determined by the source and drain boundary conditions.

Thank you!